

CardioRhythm AI — Predicting Cardiac Events Before They Happen

Series A Pitch Deck — Q2 2026.

A Software-as-a-Medical-Device platform for clinical decision support in cardiology. Headquartered in Bengaluru. Team size: 12.

Sudden cardiac death affects 4.5M Indians yearly. 80% are predictable.

- Indian ICUs and step-down wards rely on episodic ECG snapshots, missing the 6-hour pre-event window.
- CardioRhythm AI analyses continuous ECG waveforms streamed from bedside monitors and wearables.
- Surfaces clinically actionable risk windows 6 hours before an event with 94% sensitivity.
- Delivers concise risk briefs into the cardiologist's existing workflow.

CardioRhythm AI (model CRA-100) — a Class C SaMD.

- Software-as-a-Medical-Device (SaMD). Non-sterile, software-only — no physical manufacturing footprint.
- Hospital inpatient deployment + remote monitoring for at-home patients. Mixed setting of use.
- Patient population: adults 50+ at risk of cardiac events.
- User population: cardiologists and ICU staff.
- Cloud-deployed inference on AWS Mumbai; on-premise hospital connector for HIS/PACS integration.

Clinical decision support — not autonomous diagnosis.

CardioRhythm AI is intended to support clinical decision-making by cardiologists and qualified ICU staff. It DOES NOT perform autonomous diagnosis. The platform surfaces predicted risk windows and recommended interventions; the clinician interprets and decides. The cardiologist remains the responsible decision-maker for every recommendation. Not intended for paediatric use or first-line emergency triage.

Adaptive AI/ML SaMD — trained on 2M+ hours of ECG data.

- Feature extraction: morphology, rhythm variability, ST-segment trend, HRV features.
- Ensemble of gradient-boosted models + temporal CNN for sequence-level prediction.
- Adaptive — model retrains weekly on newly-labeled events. Drift monitoring built in.
- Identifiable patient ECG data processed under DPDP-aligned consent + CERT-In Safe-to-Host controls.
- Algorithm Change Protocol (ACP) to be filed alongside MD-7 per Oct 2025 CDSCO SaMD draft.

Pilot study at AIIMS Delhi — 200 patients enrolled. Published study pending.

- Sensitivity: 94% on held-out validation cohort. Specificity: 89%.
- Median lead-time: 4.2 hours before clinically-significant cardiac event.
- Pilot data submitted to IEEE EMBS for peer review.
- Multi-centre prospective study planned for 2026 across 10 hospitals.

AliveCor KardiaMobile is closest predicate. Targeting CDSCO MD-7.

- Predicate: AliveCor KardiaMobile (FDA 510(k) cleared, K181823) — single-lead ECG arrhythmia detection.
- CardioRhythm AI extends predicate materially via 6-hour prediction — hybrid pathway expected.
- Targeted CDSCO class: Class C SaMD per IMDRF risk matrix (serious clinical state × drives clinical management).
- Filing form: MD-7 (Class C/D manufacturing license, Central Licensing Authority).
- DPDP Significant Data Fiduciary registration planned once user scale crosses 10 lakh.

ISO 13485 — in progress with BSI India. Stage 1 audit Q3 2026.

Engaged with BSI India for ISO 13485:2016 certification. Stage 1 audit scheduled for Q3 2026. Internal QMS implementation under way. IEC 62304 software lifecycle classification: Class C (death or serious injury possible). Risk management per ISO 14971. Cybersecurity controls aligned to IT-Act 2000 + DPDP 2023. Two pending Indian patents on the morphology-rhythm fusion feature extractor.

12-person team. 3 active hospital pilots. 200 patients enrolled.

- Dr Arjun Mehta — Cardiologist, AIIMS Delhi. Clinical principal investigator on the pilot study.
- Dr Priya Iyer — ML/AI lead. PhD from IISc Bengaluru.
- Rohit Kumar — CTO. 14 years at Philips Healthcare on regulated medical devices.
- Pilot deployments live: AIIMS Delhi, Apollo Bengaluru, Manipal Hospital.
- LOIs from 4 additional tertiary-care hospitals contingent on MD-7 approval.

Raising Series A — INR 35Cr — to scale validation + regulatory clearance.

- Use of funds: clinical validation across 10 hospitals, MD-7 filing, ISO 13485 close-out.
- 18-month milestone: CDSCO MD-7 manufacturing license approved.
- 30-month milestone: 50,000 monitored patient-hours, 5 commercial customers.
- Contact: arjun@cardiorhythm.health · +91 80 4567 8910